



BARAN YALCIN

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WORK EXPERIENCE

Software Developer (Remote)

Freelancer

Jan. 2021 –

Izmir, Turkey

- Project or term based software (automation/embedded/desktop/web/mobile) and hardware (electronics/optics) development services for customers in various industries

Embedded Software Engineer

Trinamics

Dec. 2018 – Aug. 2020

Eindhoven, The Netherlands

- Contractor for [DAF Trucks](#), [Ibeo Automotive](#) and [ASML](#)

Systems Design Engineer

Vestel

Aug. 2016 – Dec. 2018

Manisa, Turkey

- Android camera development

Assistant

Koc University

Sep. 2013 – Sep. 2015

Istanbul, Turkey

- Research assistant at [KirazLab](#)
- Teaching assistant for Microprocessors, Electronic Design and Semiconductor Devices courses

PROJECTS

QA Automation

using Python, TensorFlow

Appsilon

Training and integration of CNN model for determining the quality, based on the image of a diamond seed.

Thermal Imaging Algorithm

using Java, ImageJ, Python, OpenCV

Appsilon

Development of an algorithm for segmenting and predicting the temperature of diamond seeds, throughout the whole process by acquiring images.

AI model integration

using Python, TensorFlow, Docker, Java, Spring

[Response Ortho](#)

Dockerized and integrated an existing CNN model to a web application.

Motor controller application

using JavaScript, React Native, C++

[Teknofil](#)

Bluetooth based Android application that communicates with an ESP NodeMCU for controlling a motor.

ImageJ camera plugin

using Java, JNI, C++

Appsilon

Design of an optical imaging system and implementation of a ImageJ plugin for capturing periodical images from Basler's acA2500-14um and acA2500-14uc cameras.

Optical emission spectroscopy data logger/visualizer

using C++, Qt, Qwt

Appsilon

30+ days storage capable data logger/visualizer for Ocean Optics' HR2000+ device.

Laser position sensor data logger/visualizer

Appsilon

using C++, Qt, Qwt

30+ days storage capable data logger/visualizer for Acuity's AR500 device.

Automated UI

Ibeo Automotive

using Python, PyQt

File structure agnostic UI for editing XML based simulation scenario files.

Scenario variation module for the LIDAR sensor model simulator

Ibeo Automotive

using Python

Automatic generation of new scenario files given a variation and a base scenario file.

DAF MY21

DAF Trucks

using Simulink, Rhapsody, Python, MATLAB

Development of model based embedded software for non-engine related functionalities of the vehicle together with the VECU team, following V-Model and Scrum methodologies.

"Hey DAF!"

DAF Trucks

using Java, cmusphinx, RPi, PiCAN2, Python

Android app for interacting with the vehicle using voice commands.

readerx

using C, X11, Poppler, Cairo and GLib

Open source, lightweight PDF reader with Vim-like keybindings.

Age and gender predictor

Vestel

using C++, Caffe, Snapdragon Neural Processing Engine

CNN based vision algorithm (in Android HAL) that predicts age and gender of the subject(s) in the viewfinder.

Smileshot

Vestel

using C++

AI based vision algorithm (in Android HAL) that takes a snapshot when a smile is detected in the viewfinder.

Vision algorithm integration

Vestel

using C++

Integration of third-party vision algorithms (into Android HAL) such as bokeh, beauty mode, low light shot, HDR and panorama stitching.

ctools

Vestel

using Batch script

Tool for compiling and flashing ISP header files into a Qualcomm based Android smartphone.

OIS test bench

Vestel

using C

An apparatus which simulates the hand tremor effect for measuring the efficacy of Optical Image Stabilization algorithm.

ISP and 3A algorithm tuning

Vestel

using C, Chromatix, MATLAB, Imatest

Image processing algorithm tuning for Qualcomm based Android smartphones.

Konfokal

KirazLab

using Python, PyQt, SciPy, Numpy, matplotlib, PIL, pyserial

Supervisory program with GUI for interacting with the CDAQ unit through USB. It can capture and process an image based on user's scanning preferences.

CDAQ

KirazLab

using C, Eagle, TI Tiva

Hardware and firmware development of a CDAQ (Control and Data Acquisition) unit for the Laser Scanning Confocal Microscope. It consists of an ADC for reading the photomultiplier tube, and DACs for driving the galvanometer scanning mirrors.

Linear programming on GPU

Koc University

using CUDA C

Implementation of revised simplex algorithm on a NVIDIA GPU as the final year project.

EDUCATION

Master of Science | *Optoelectronics & Photonics Engineering*

Sep. 2013 – Sep. 2015

Koc University

Istanbul, Turkey

- Thesis: Hardware/Software Design and Implementation of a Laser Scanning Confocal Microscope Controller Using Open Design Approach

Bachelor of Science | *Electrical & Electronics Engineering*

Sep. 2009 – Jun. 2013

Koc University

Istanbul, Turkey

SKILLS

Languages: C, C++, JavaScript, Java, Python, bash, MATLAB, Simulink, Assembly, Forth, batch, UML

Tools: gcc, MSVC, gdb, valgrind, llvm, make, meson, cmake, git, SVN, Jenkins, n8n

Equipment: Oscilloscope, function generator, JTAG debugger, logic analyzer, multimeter, soldering iron